LINE SEGMENTS AND INTERCEPTS

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Line Segments

- Distance Formula
- Division of Line Segment
- Midpoint of Line Segment

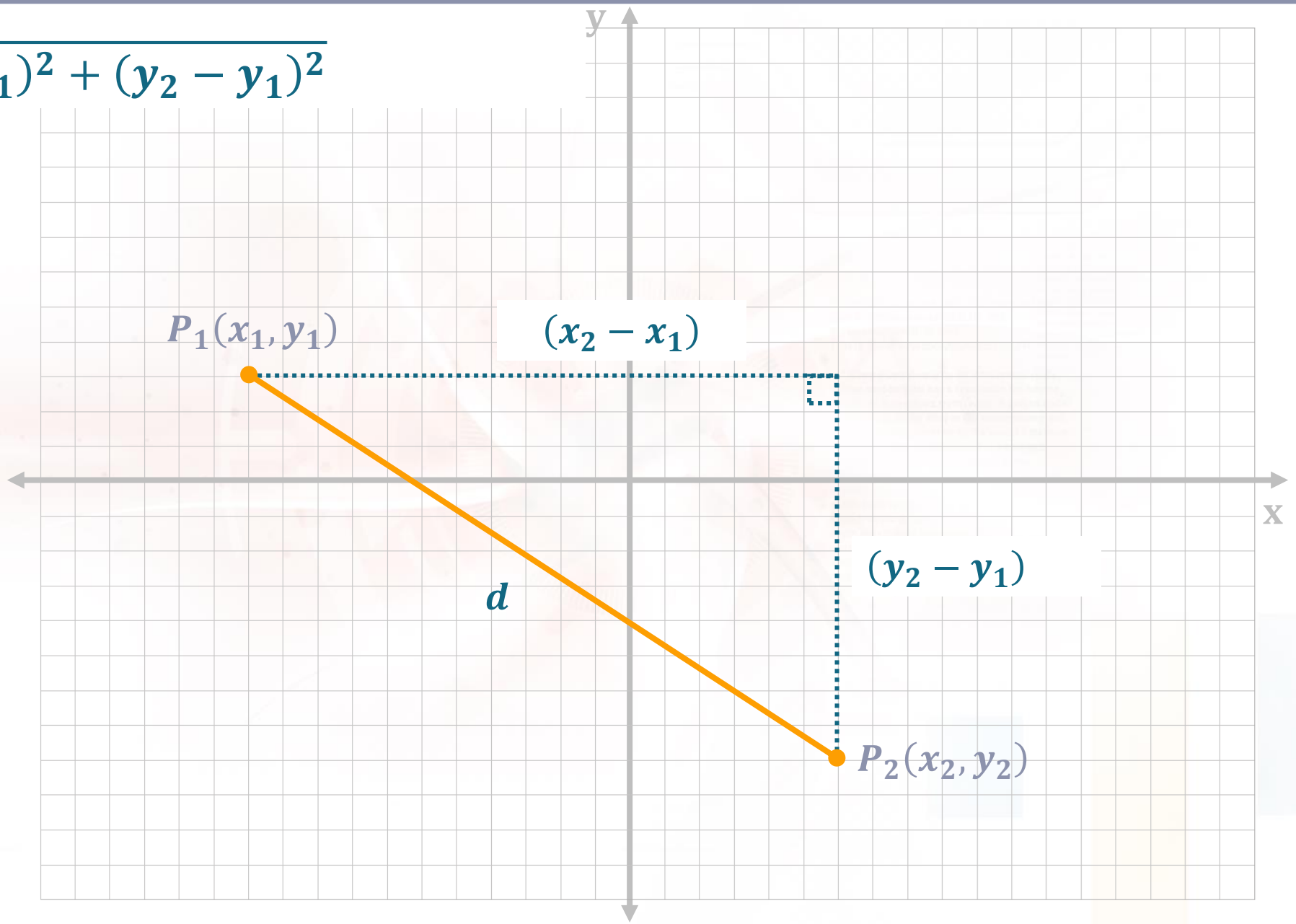
Intercepts of the Graph of an Equation

- x-intercepts
- y-intercepts

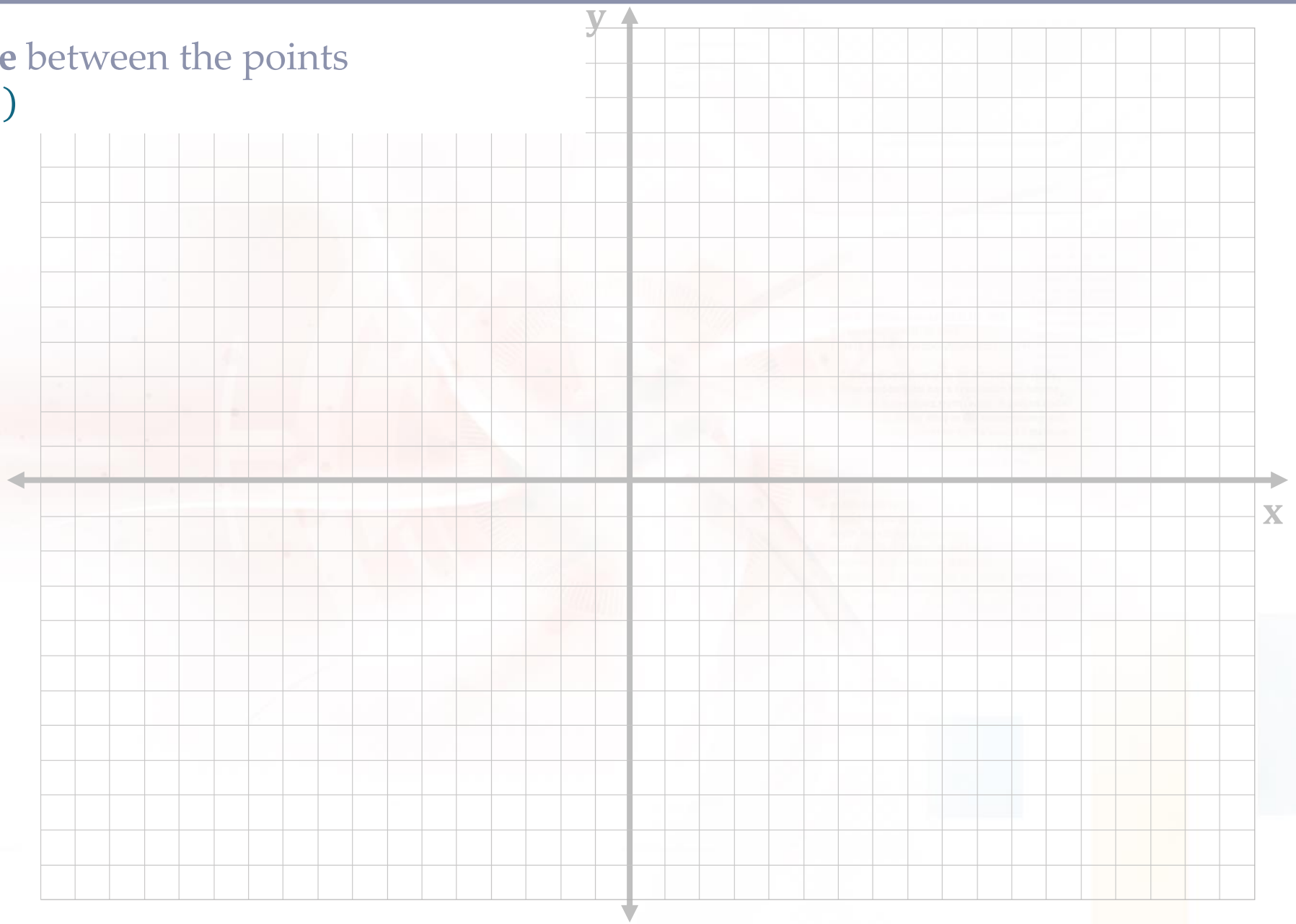
LINE SEGMENTS

THE DISTANCE FORMULA

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



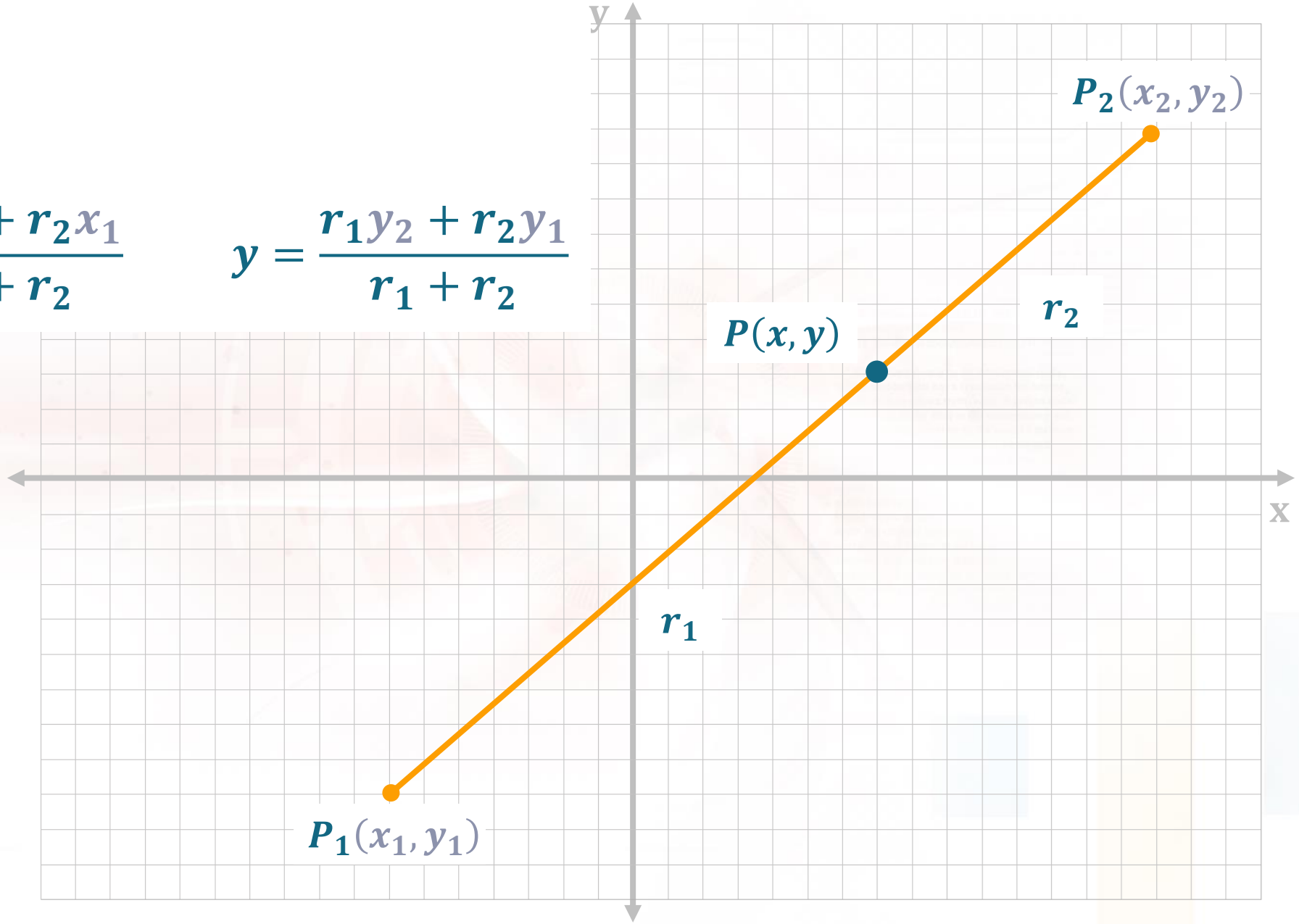
Find the **distance** between the points $(4, -2)$ and $(6, 5)$



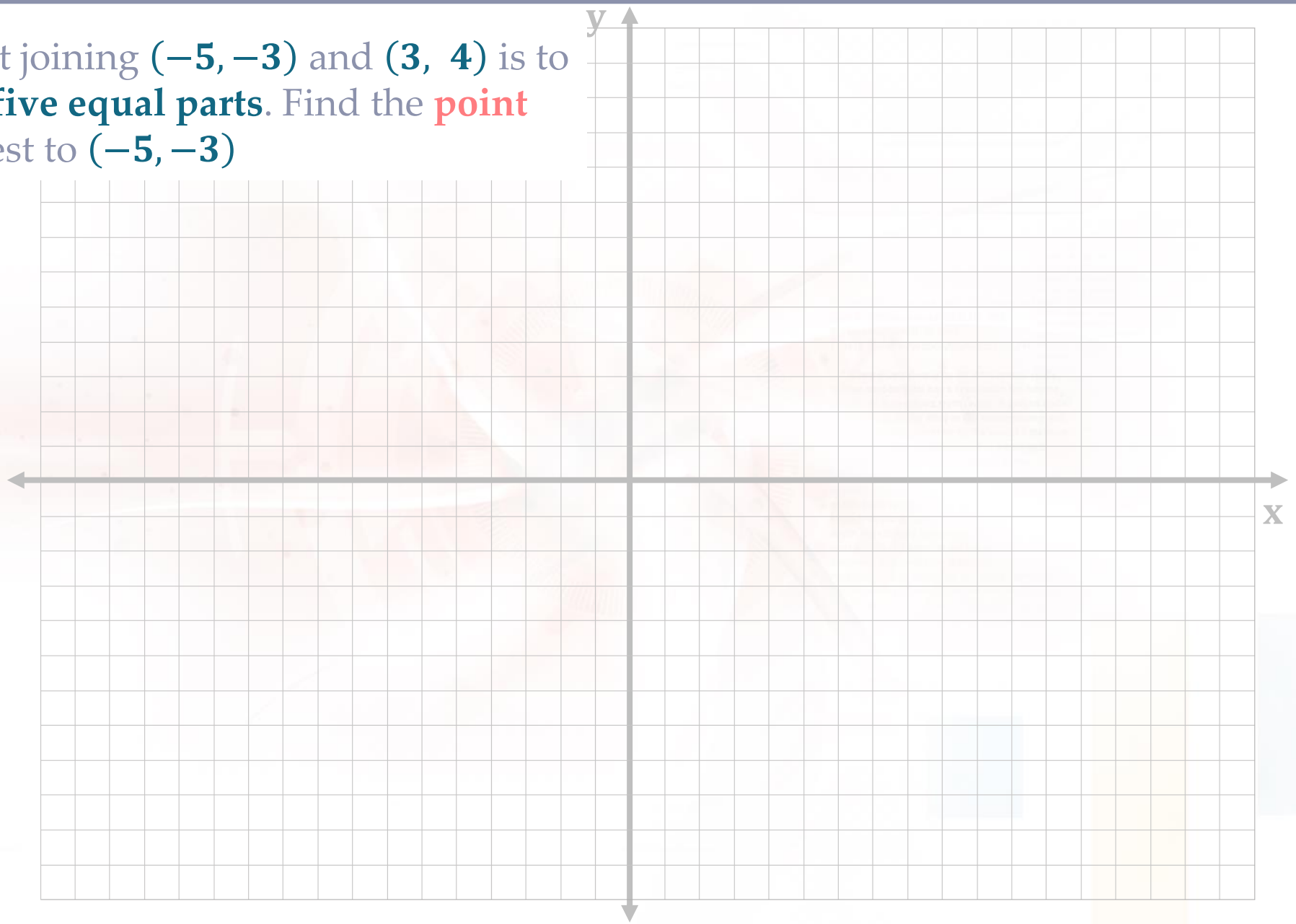
THE DIVISION OF LINE SEGMENT

$$\frac{P_1P}{PP_2} = \frac{r_1}{r_2}$$

$$x = \frac{r_1x_2 + r_2x_1}{r_1 + r_2} \qquad y = \frac{r_1y_2 + r_2y_1}{r_1 + r_2}$$



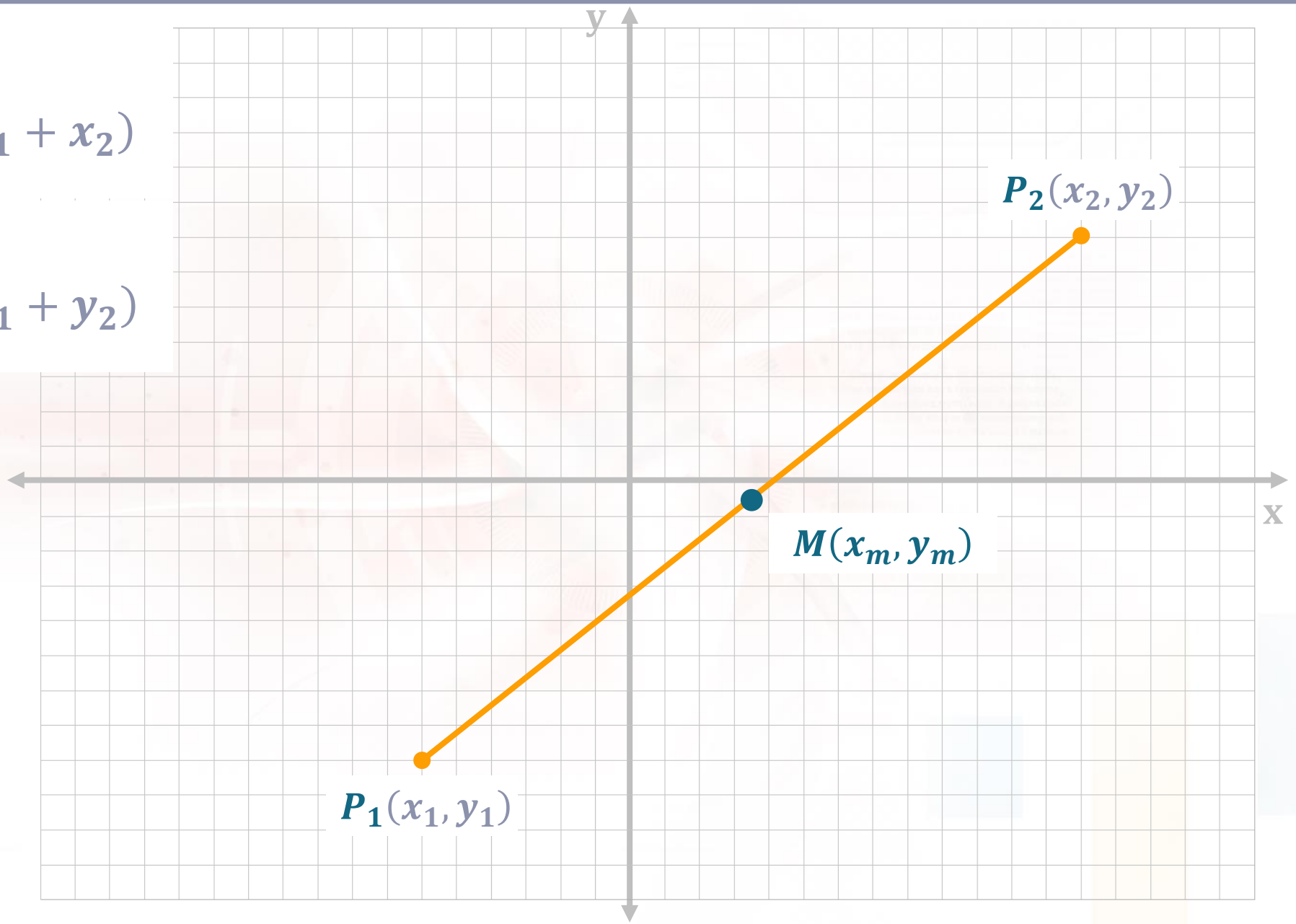
The line segment joining $(-5, -3)$ and $(3,\ 4)$ is to be divided into **five equal parts**. Find the **point of division** closest to $(-5, -3)$



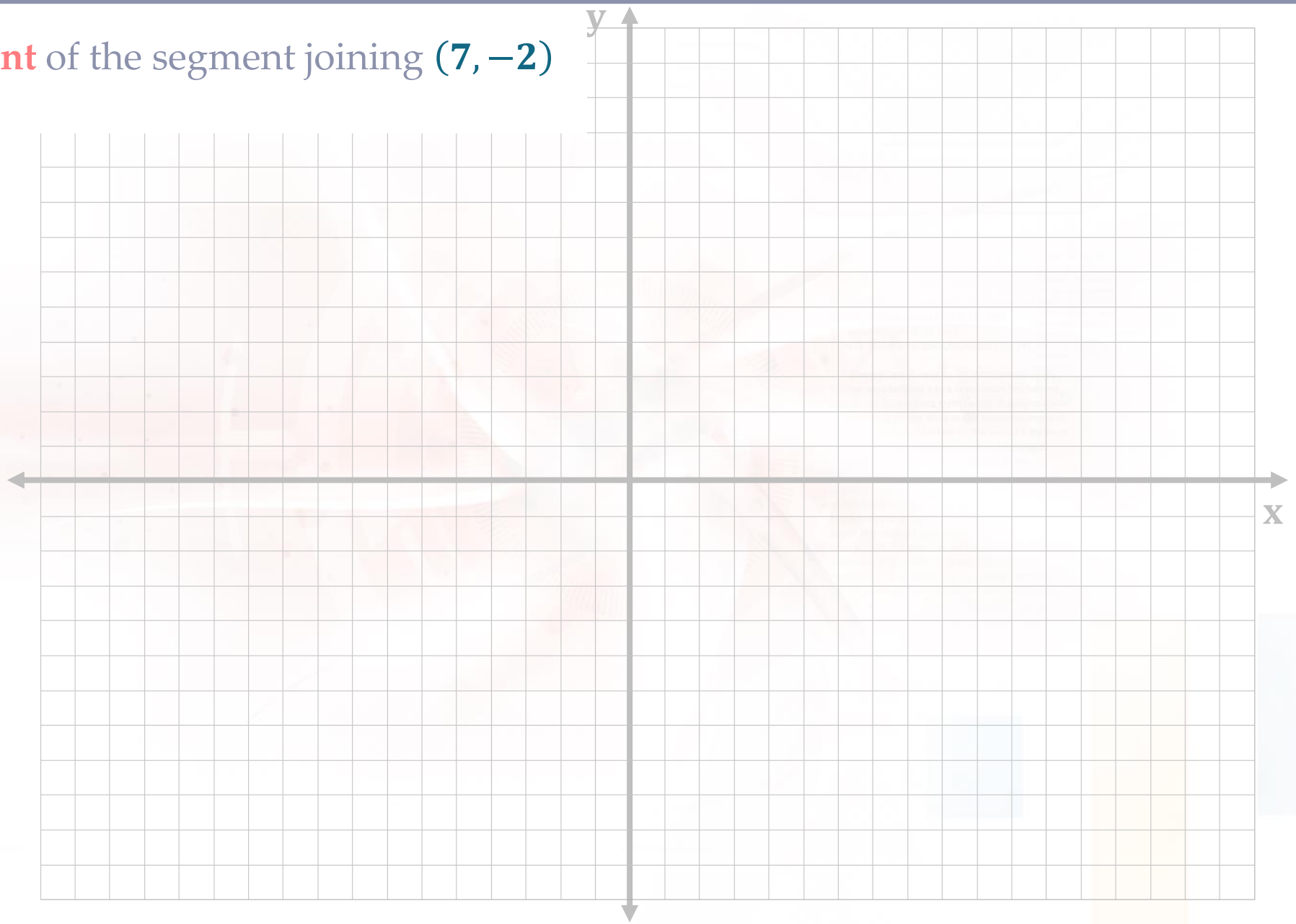
MIDPOINT OF A LINE SEGMENT

$$x_m = \frac{1}{2}(x_1 + x_2)$$

$$y_m = \frac{1}{2}(y_1 + y_2)$$



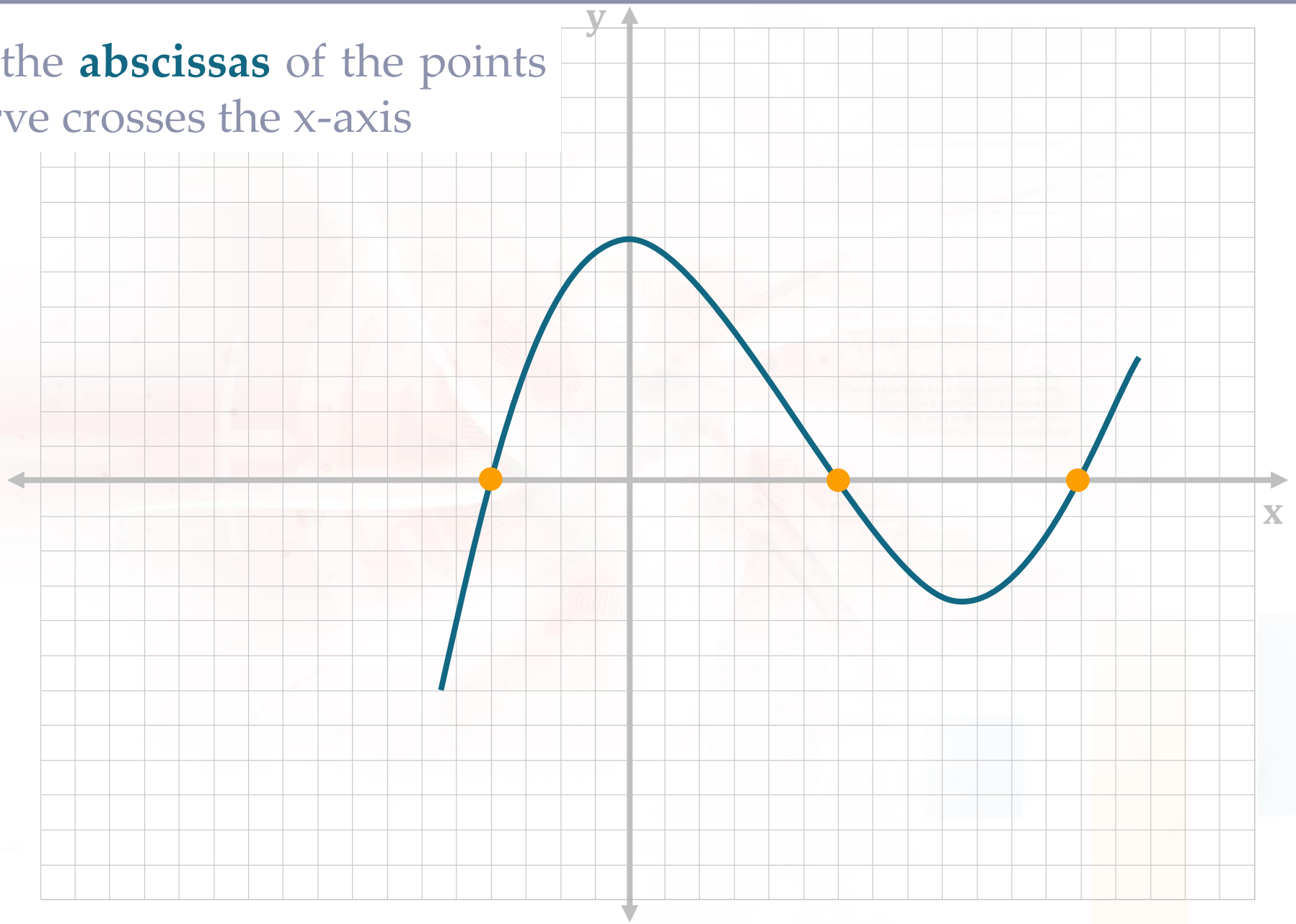
Find the **midpoint** of the segment joining $(7, -2)$ and $(-3, 5)$



INTERCEPTS OF THE GRAPH OF AN EQUATION

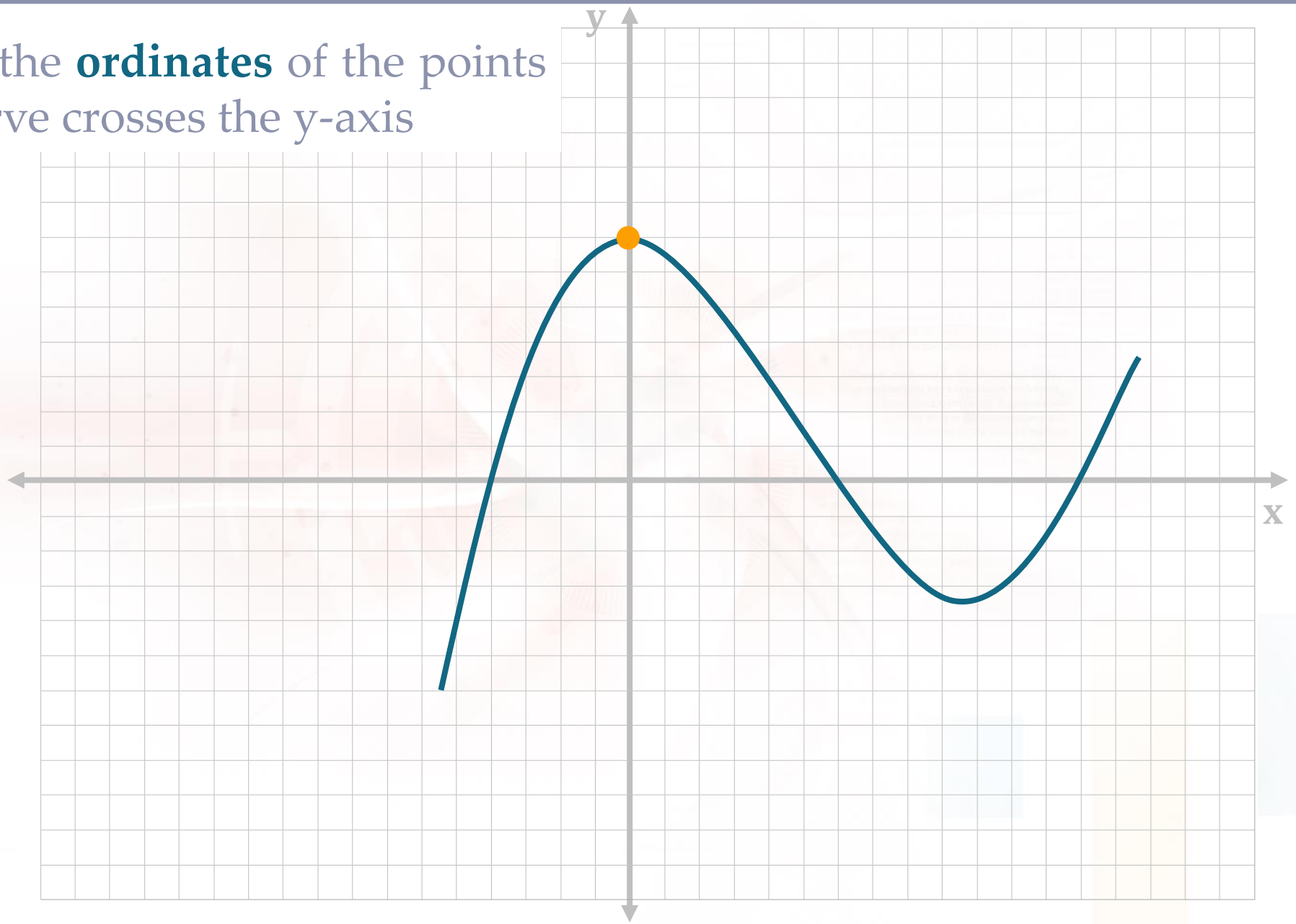
$x - intercepts$

the values of the **abscissas** of the points where the curve crosses the x-axis

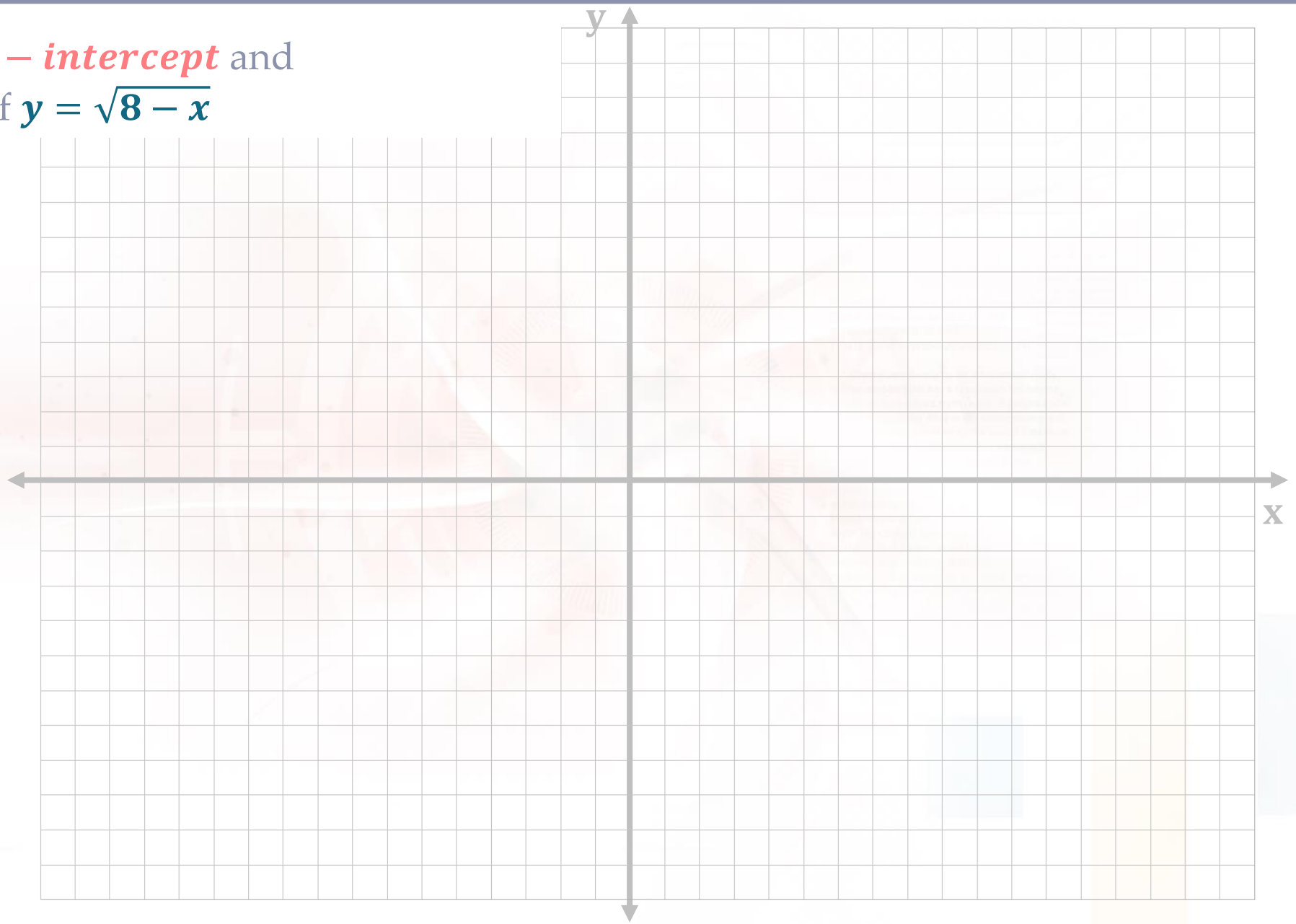


$y - intercepts$

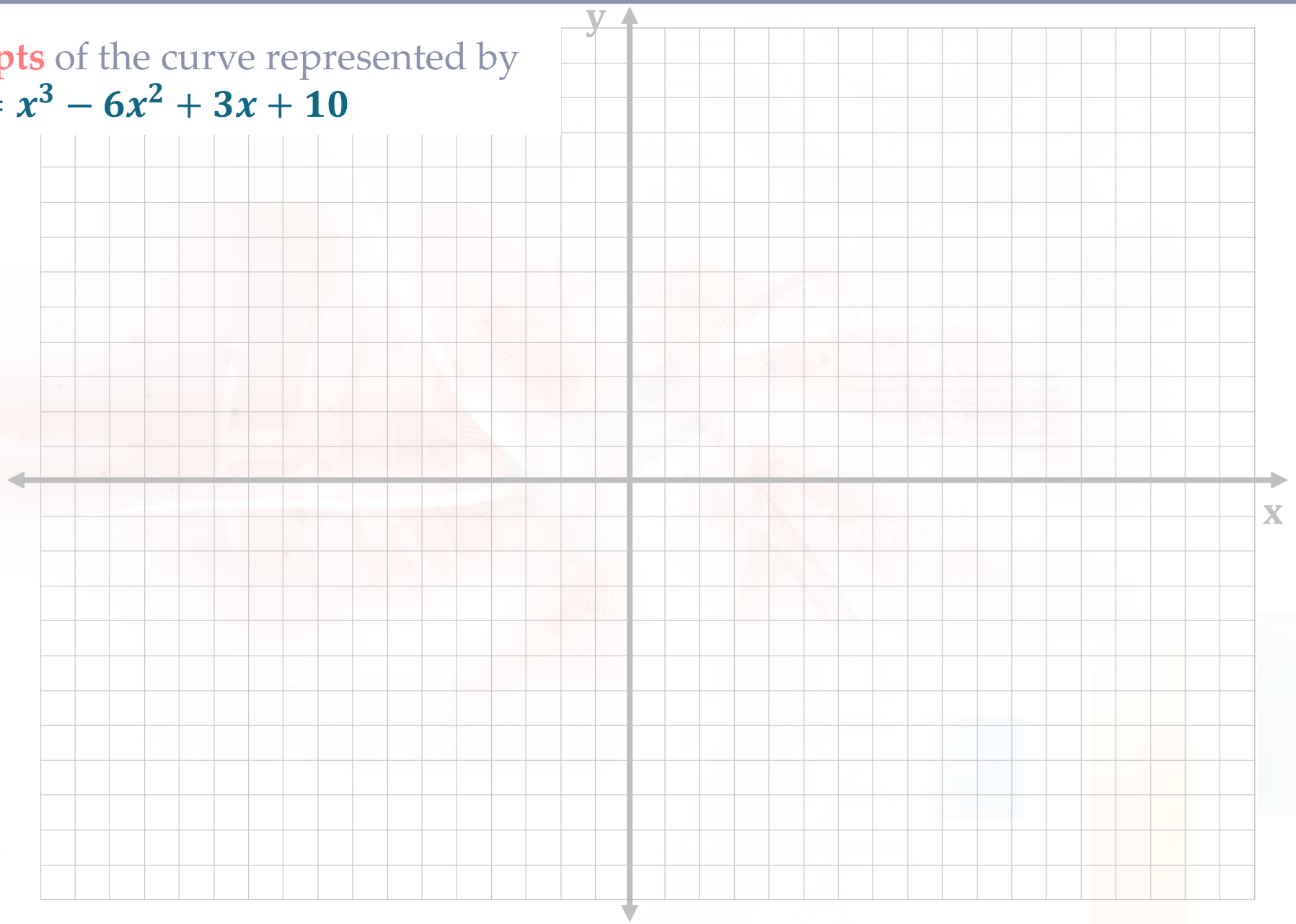
the values of the **ordinates** of the points where the curve crosses the y-axis



Determine the $x - intercept$ and $y - intercept$ of $y = \sqrt{8 - x}$



Find the **intercepts** of the curve represented by the equation $\boldsymbol{y = x^3 - 6x^2 + 3x + 10}$



SEATWORK

1. Find all the **points** on the **x-axis** that are a **distance 5** from $\boldsymbol{P(-2,4)}$.

2. Find the **point** with coordinates of the form $\boldsymbol{(2a,a)}$ that is in the third quadrant and is a **distance 5** from $\mathbf{P(1,3)}$.

3. Find the **intercepts** of the curve represented by the equation

$$x^2y + 2xy - x - 3y + 3 = 0.$$